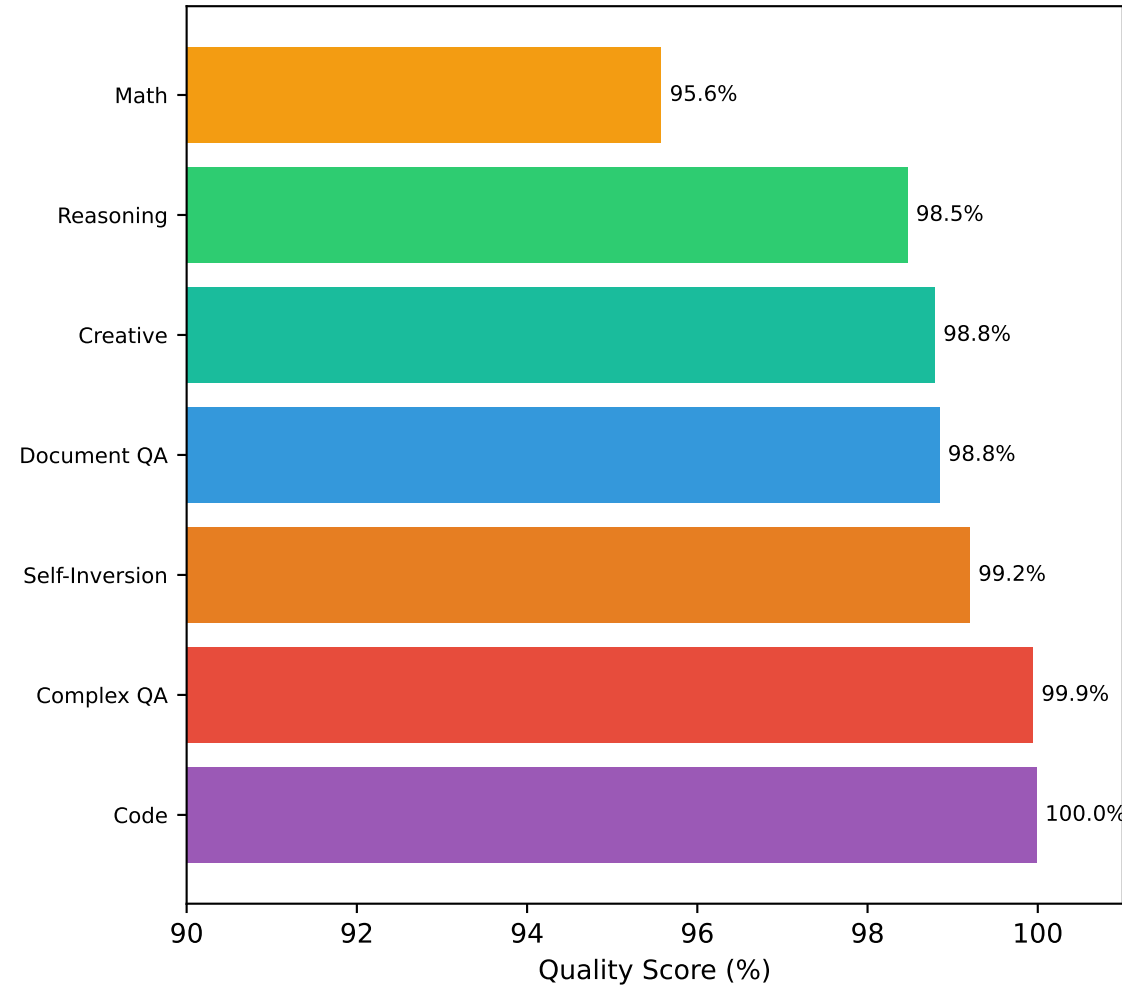
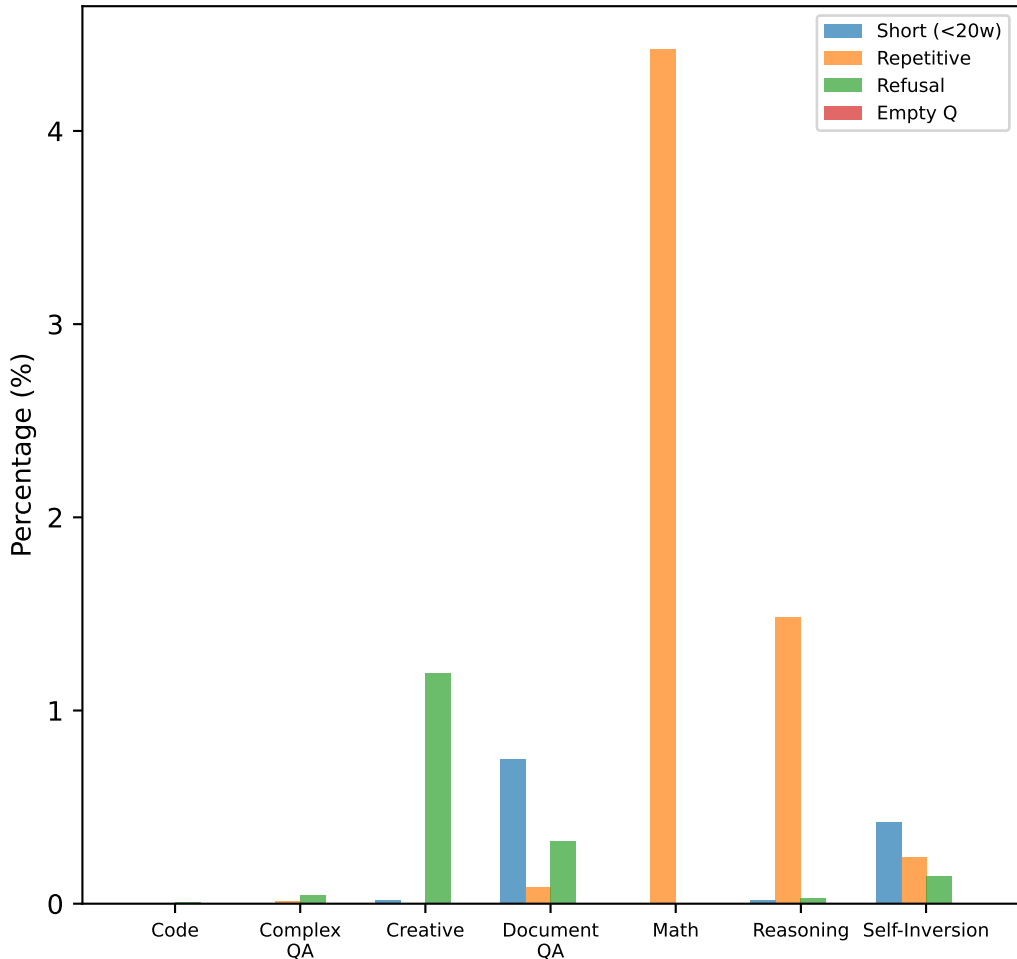


Fig 166: Data Quality Scorecard

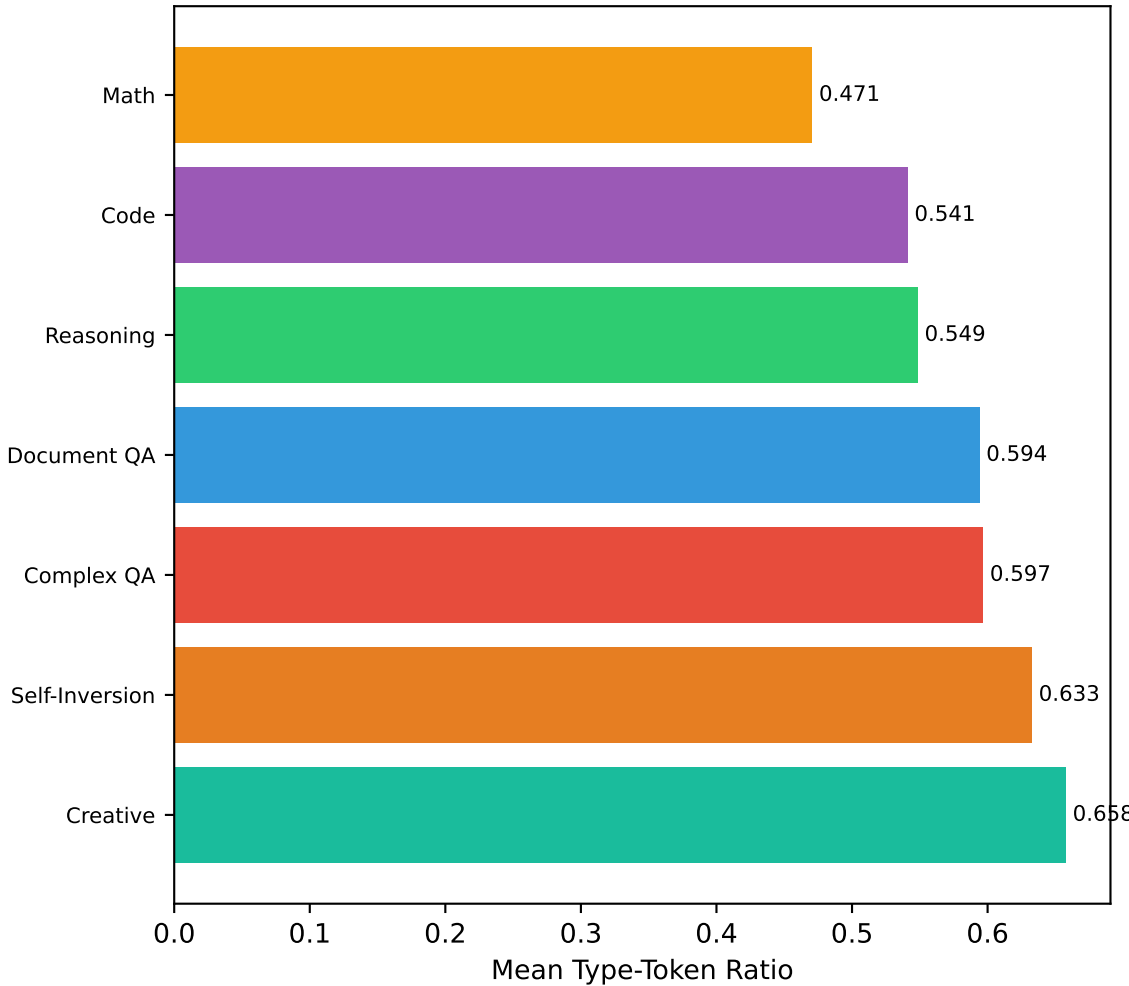
(a) Overall Quality Score



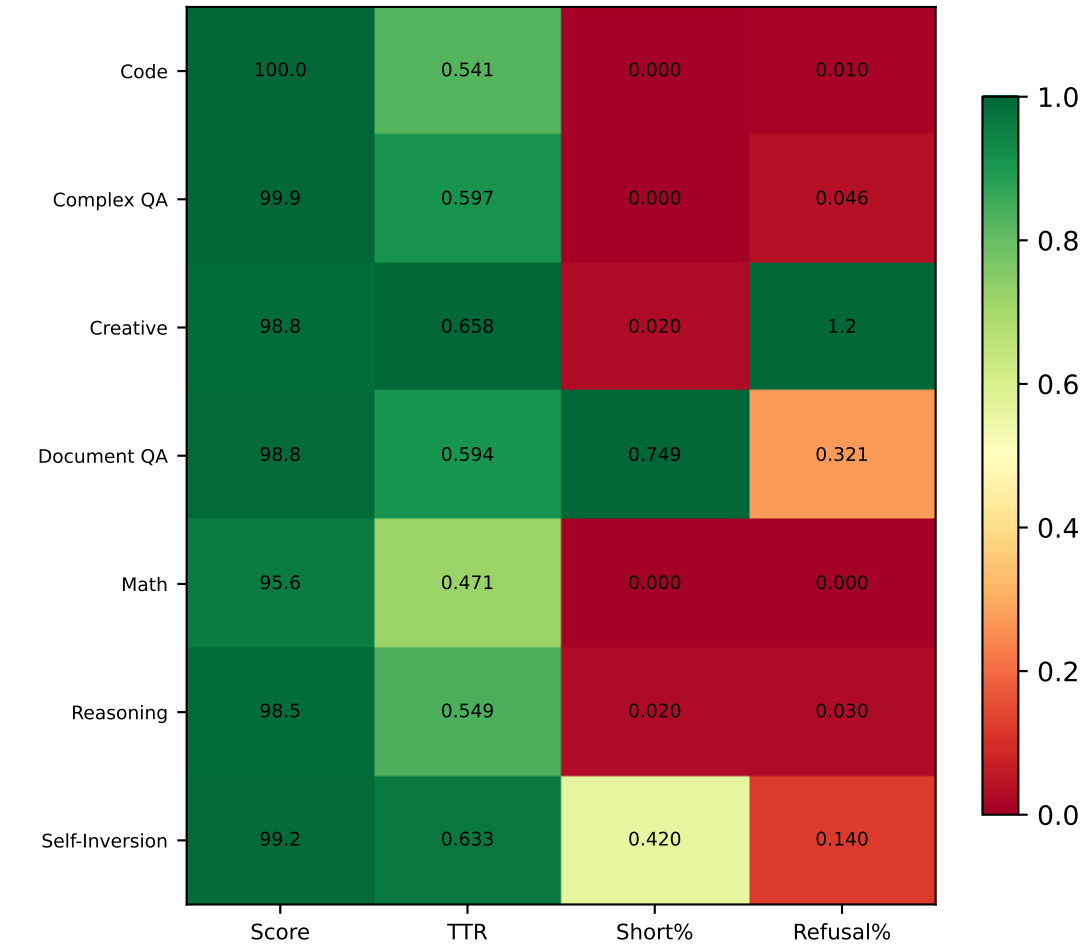
(b) Issue Breakdown



(c) Lexical Diversity (TTR)



(d) Quality Heatmap



(e) Quality Grades

| Pipeline       | Score  | TTR   | Issues | Grade |
|----------------|--------|-------|--------|-------|
| Code           | 100.0% | 0.541 | 0.0%   | A+    |
| Complex QA     | 99.9%  | 0.597 | 0.1%   | A+    |
| Creative       | 98.8%  | 0.658 | 1.2%   | A     |
| Document QA    | 98.8%  | 0.594 | 1.2%   | A     |
| Math           | 95.6%  | 0.471 | 4.4%   | A-    |
| Reasoning      | 98.5%  | 0.549 | 1.5%   | A     |
| Self-Inversion | 99.2%  | 0.633 | 0.8%   | A+    |

(f) Verdict

Quality Scorecard Summary

All pipelines score >90%  
Mean quality: 98.7%  
Mean TTR: 0.577

Best quality: Code (100.0%)  
Best TTR: Creative (0.658)

Total refusals across all: 235  
Total short answers: 209

Verdict: PUBLICATION QUALITY  
All pipelines pass quality threshold